

LESSON 3.4

WRITING SMALL NUMBERS IN SCIENTIFIC NOTATION



LESSON INTRODUCTION

MA.8.NSO.1.4 Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.

LEARNING TARGETS

- I can express small numbers in scientific notation.
- I can determine how many times larger or smaller a number is compared to another expressed in scientific notation.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole number exponents and rational number bases.

WORKING TOWARD

MA.9.12.NSO.1.1 Extend previous understanding of the laws of exponents to include rational number exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

ESSENTIAL QUESTIONS

- How do you express small numbers in scientific notation?
- How many times larger or smaller is the number compared to another number expressed in scientific notation?

LESSON NARRATIVE

Students explore real-world examples of small numbers and discover how to express them in scientific notation. They also discover how to determine how many times larger or smaller a number is compared to another expressed in scientific notation.

COMPONENT SUMMARY

3.4.1 WARM-UP (~5 MIN.)

Develop growth mindset

3.4.2 GUIDED INSTRUCTION (15 MIN.)

Discover how to write small numbers in scientific notation

3.4.3 GUIDED PRACTICE (5 MIN.)

Practice writing small numbers in scientific notation

3.4.4 COLLABORATION (10 MIN.)

Compare small numbers written in scientific notation

3.4.5 YOUR TURN! (5 MIN.)

Practice writing and comparing small numbers in scientific notation

3.4.6 WRAP-UP (~5 MIN.)

Match scientific notation to standard or word form

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Scientific Notation
- Standard Form

MATERIALS

- Growth Mindset video
- (Optional) Highlighters or Colored Pencils
- (Optional) Cards for Wrap-Up

ADDITIONAL PREPARATION

- Warm-Up 3.4.1 contains a video to accompany the question prompts.
- Consider structuring the 3.4.6 Wrap-Up as a card sort. Prepare cards before lesson.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may incorrectly think that the difference in the powers of 10 is how many times larger or smaller a number is than another number. For example, they may think 10^{-5} is two times larger than 10^{-3} , because $(-3)-(-5)=2$, instead of 10^2 or 100 times larger.

THINGS TO CONSIDER

- Encourage students to check their work by expressing the scientific notation in standard form or vice versa.
- Consider reviewing place values less than one.
- Consider creating an anchor chart with the form of scientific notation as $a \times 10^n$ where $1 \leq |a| < 10$. Numbers can be expressed using powers of ten that are not in this form but the definition of scientific notation is that the absolute value of a should be at least 1 and less than 10. Consider using both the mathematical representation of the constraint on a as well as the written description of the constraint.
- Consider reviewing the names of place values of the decimal representations of negative powers of 10 or providing them on an anchor chart.
- If time is an issue, then consider assigning 3.4.5 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Turn and Talk

Prompt students to take turns during 3.4.4 Collaboration. Have students share their strategies and answers. Then, ask the partner to share if they agree or disagree and why.

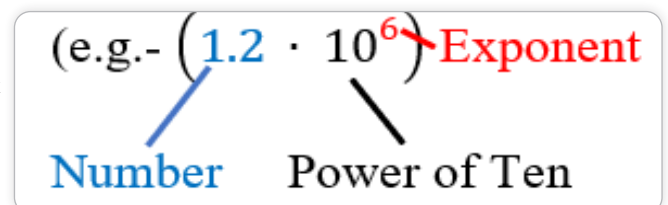
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

In 3.4.2 Guided Instruction students explore and discover how to use multiple representations to express small numbers. Engage students in discussions about how the scientific notation is connected to the standard form of the number.

DIFFERENTIATE INSTRUCTION

In 3.4.2 Guided Instruction, provide a visual entry point by providing students with a template to use when expressing numbers in scientific notation. In 3.4.4 Collaboration, provide an auditory entry point by prompting students to read their completed statements in questions 1b-1e aloud.



3.4.1 WARM-UP: MATH MINDSET

DIFFERENTIATE INSTRUCTION

Consider providing students with sentence starters to aid them in constructing their responses to the writing prompts.



3.4.1 Warm-Up: Math Mindset

In the story of *The Tortoise and the Hare*, the hare had a fixed mindset while the tortoise had a growth mindset.

1. "Failure is an almost essential step to success."
Describe a time in which you experienced failure before reaching success.

Answers will vary. I had a hard time learning to do long division but kept trying until I got it.

2. What can you do to overcome the thoughts of, "I'm not good at _____." or "I can't _____."?

Answers will vary. Remind myself of all the things I can do. Remind myself of all the times I have tried until I figured out something.



3.4.2 Guided Instruction: Writing Small Numbers in Scientific Notation

Powers of ten can also be used to express small numbers in scientific notation.

1. A single red blood cell is about 0.000007 meters in diameter.
 - a. In what place is the 7 in the value, 0.000007?
millionths
 - b. Which power of 10 relates to this place value?
 10^{-6}
 - c. Express 0.000007 in scientific notation.
 7×10^{-6}
2. Consider the value 0.0000672.
 - a. What value between 1 and 10 could be multiplied by a power of 10 to result in 0.0000672?
6.72
 - b. Which digit is in the ones place of your answer to Part A?
6
 - c. In which place value is that digit in 0.0000672?
hundred thousandths
 - d. Which power of 10 relates to this place value?
 10^{-5}
 - e. Express 0.0000672 in scientific notation.
 6.72×10^{-5}
3. Consider the value 3.4×10^{-9} .
 - a. Complete the statements.
 10^{-9} relates to the *billionths* place. When rewriting 3.4×10^{-9} in standard form, the *3* will be in the billionths place, followed by the *4*. The places before the 3 will each hold the digit, *0*.
 - b. Express this value in standard form.
0.0000000034

3.4.2 GUIDED INSTRUCTION: WRITING SMALL NUMBERS IN SCIENTIFIC NOTATION

TEACHER MOVES

The lesson contains many interesting facts. Engage students in those facts by asking questions such as:

- How do you think they determined that measurement?
- Which do you think is harder to measure – really small things or really large things?

DIFFERENTIATE INSTRUCTION

Provide a visual entry point by providing students with a template to use when expressing numbers in scientific notation.

TEACHER MOVES

Students continue to explore and discover how to use multiple representations to express small numbers. Engage students in discussions about how scientific notation is connected to the standard form of the number. Rewriting numbers in scientific notation should not be taught by moving decimal places. Rather, in the context of place value and powers of 10 as built-in previous lessons. The Guided Instruction is set up to continue using place value.

3.4.3 GUIDED PRACTICE: WRITING SMALL NUMBERS IN SCIENTIFIC NOTATION

TEACHER MOVES

Encourage students to check their work by expressing the scientific notation in standard form or vice versa.

ENRICHMENT

How does writing small numbers in scientific notation compare to writing large numbers in scientific notation? What are the similarities and differences?



3.4.3 Guided Practice: Writing Small Numbers in Scientific Notation

1. The density of oxygen is 1332 millionths of a gram per cubic centimeter.
 - a. Express this value as a decimal in standard form.
 0.001332
 - b. Express this value in scientific notation.
 1.332×10^{-3}
2. The Mycoplasmas bacteria is about 2.5×10^{-4} millimeter (mm) in size.
Express this value in standard form.
 0.00025



3.4.4 Collaboration: Comparing Small Numbers Written in Scientific Notation

1. Work with your partner to complete the following.

a. Complete the table.

Scientific Notation	Standard Form
7.2×10^{-4}	0.00072
1.8×10^{-7}	0.00000018

b. Complete the statement to compare the factors being multiplied by powers of 10 in each number that is expressed in scientific notation.

7.2 is 4 times ☐ smaller ☒ larger than 1.8.

c. Complete the statement to compare the powers of 10 in each number that is expressed in scientific notation.

10^{-4} is 1000 times ☐ smaller ☒ larger than 10^{-7} .

d. Complete the statement.

The value 7.2×10^{-4} is 4000 times ☐ smaller ☒ larger than 0.00000018.

3.4.4 COLLABORATION: COMPARING SMALL NUMBERS WRITTEN IN SCIENTIFIC NOTATION

TEACHER MOVES

Turn and Talk

Prompt students to take turns during this component. Have students share their strategies and answers. Then, ask the partner to share if they agree or disagree and why.

DIFFERENTIATE INSTRUCTION

Provide an auditory entry point by prompting students to read their completed statements in questions 1b-1d aloud.

3.4.5 YOUR TURN!

DIFFERENTIATE INSTRUCTION

Allow informal observation throughout the first two activities to dictate how best to use this component. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.

TEACHER MOVES

Use the real-world examples to prompt discussions about the utility of expressing small numbers in scientific notation.



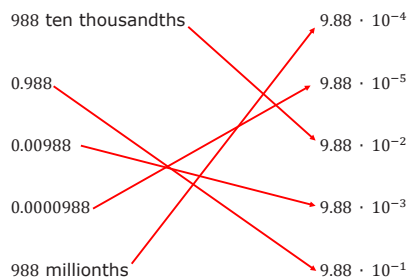
3.4.5 Your Turn!

1. Express each number in scientific notation.
 - a. The diameter of a grain of sand is 0.0024 inch.
 2.4×10^{-3}
 - b. The mass of a dust particle is about 753 trillionths of a kilogram (kg).
 7.53×10^{-10}
2. Express each number in standard form.
 - a. 3×10^{-8}
0.00000003
 - b. 1.2379×10^{-1}
0.12379



3.4.6 Wrap-Up: Ticket Out the Door

1. Match each value on the left with its expression in scientific notation on the right by drawing lines to connect the equivalent expressions.



3.4.6 WRAP-UP: TICKET OUT THE DOOR

DIFFERENTIATE INSTRUCTION

Consider providing a kinesthetic entry point by structuring this component as a card sort.

QUESTIONING

Challenge students who finish early to research real-world examples of very small numbers and determine how many times larger or smaller a number written in scientific notation is than another number written in scientific notation.